

Flutter Assessment & Training Portal

Master Operational Guide for Teachers, Reviewers & Administrators

1. System Overview & Architecture

The Flutter Assessment & Training Portal is an advanced, Docker-isolated web platform designed to evaluate students across progressive stages of Dart and Flutter development. It provides real-time interactive UI previews, multi-file project support, and hybrid automated/manual grading.

- **Dockerized Execution Engine:** Every student submission runs inside an isolated Docker container (flutter-runner) with pre-warmed Flutter SDKs and Linux font bundles.
- **Multi-File Workspace Support:** Students and teachers can work across multi-file project structures (e.g., lib/main.dart, lib/models/item.dart, lib/widgets/card.jsx). Code is stored and serialized as JSON file maps.
- **Real-Time Interactive Previews:** For UI challenges, the portal compiles Flutter Web bundles on the fly, rendering live interactive applications directly inside iframe viewports.
- **Extended Proxy Timeouts (300s):** The Nginx proxy layer is configured with 5-minute timeouts to ensure complex Flutter Web builds complete without dropping connections.

2. Level Creation & Progression Management

The curriculum is structured into 7 progressive stages. Students must clear earlier levels before advancing to complex state management and full-stack project builds.

- **Level 1A–1C (Dart Fundamentals):** Pure Dart, Console Output, Variables, OOP, Null Safety, Collections.
- **Level 2A–2C (Flutter Basics):** UI Construction, Scaffold, Row, Column, Stack, Images, Icons (No State/Networking).
- **Level 3A–3C (Interactive Flutter):** StatefulWidget, TextFields, Buttons, Forms, Controllers, Navigation.
- **Level 4A–4C (State Management):** Provider, Riverpod, Shared State, Lifecycle management.
- **Level 5A–5C (Networking):** HTTP, REST APIs, JSON CRUD, Mock backend integration.
- **Level 6A–6C (Storage):** Local persistence, Shared Preferences, SQLite, File handling.
- **Level 7+ (Capstone Projects):** E-commerce, Hospital App, Banking App, Chat App.

How to Create & Configure Levels:

1. Navigate to **Levels Management** in the Admin Sidebar and click **+ Add New Level**.
2. Set the **Level Code** (e.g., 3A), **Assessment Type**, and **Title**.
3. Configure **Question Count** (number of problems assigned per session) and **Duration Minutes**.
4. Set the **Pass Threshold (%)** required to unlock the next level (default: 70% or 80%). Click **Save Level**.

3. Question Bank & Problem Management (In-Depth Guide)

The **Question Bank** (AdminQuestions_new.jsx) is where teachers design interactive challenges. Here is how to utilize all advanced capabilities:

- A. Multi-File Workspace & Starter Code:** In the Starter Code section, define boilerplate for lib/main.dart. Click **+ Add File** to create supplementary files (e.g., lib/models/user.dart), teaching modular architecture.
- B. Resource Bundles (Images, Fonts, JSON):** In the Resources tab, upload PNGs, custom TTF fonts, or datasets. The Docker runner automatically copies them to assets/images/ inside the container and auto-registers them in pubspec.yaml!
- C. Third-Party Package Injection:** In the Required Packages section, select checkboxes for libraries like provider, http, shared_preferences, google_fonts, or fl_chart. They are automatically injected into pubspec.yaml during compilation.
- D. Mock APIs & Backend Integration (Level 5+):** For networking problems without internet access, set a **Mock API Route** (e.g., /api/mock/products) and paste the exact JSON response payload. The container intercepts HTTP calls and returns this mock data!
- E. Mock Database Seeding (Level 6+):** Provide seed SQL queries or initial JSON data in the **Mock DB Seed** box to prepopulate databases before student code executes.
- F. Custom Test Code (Widget Verification):** Write automated verification suites using Flutter's testWidgets library to assert button taps, text renders, and state transitions.

4. Comprehensive Bulk Upload Guide & Templates (CSV/XLSX)

The Flutter Assessment Portal supports instant bulk importing for **Students**, **Staff Members**, **Test Slot Registrations**, and the **Question Bank** using standard CSV/Excel files. Template files are also provided inside the bulk_templates/ directory of your repository.

1. Save your spreadsheet as a **UTF-8 Encoded CSV** (.csv) or Excel Spreadsheet (.xlsx).
2. Navigate to the respective page and click **Bulk Import / Upload -> Choose File -> Upload**.
3. The backend validates headers and auto-enrolls accounts instantly.

A. Student Bulk Import (AdminUsers.jsx):

Used to batch-register student accounts. New students assigned role STUDENT automatically start unlocked at Level 1A.

Header Name	Req?	Description & Format / Example
email	YES	Unique student email (e.g., student1@bitsathy.ac.in)
full_name	YES	Student's full name (e.g., Aarav Sharma)
enrollment_no	NO	University enrollment number (e.g., 737622CS101)
roll_no	NO	Roll number / Register number (e.g., 22CS101)
role	NO	Must be STUDENT (or defaults to STUDENT)

Header Name	Req?	Description & Format / Example
auth_provider	NO	Either GOOGLE (OAuth) or LOCAL (password login). Defaults to GOOGLE
password	COND	Required only if auth_provider is LOCAL

B. Staff / Faculty Bulk Import (AdminStaff.jsx):

Used to batch-register faculty members and administrators.

Header Name	Req?	Description & Format / Example
email	YES	Faculty official email (e.g., faculty1@bitsathy.ac.in)
full_name	YES	Faculty full name (e.g., <i>Dr. Karthik S.</i>)
staff_id	NO	Staff ID / Employee Code (e.g., BITFAC001)
role	YES	Either TEACHER or ADMIN
auth_provider	NO	Either GOOGLE or LOCAL. Defaults to GOOGLE
password	COND	Required only if auth_provider is LOCAL

C. Test Slot Registration Bulk Import (AdminTestSlots.jsx):

Inside any Test Schedule, click **Bulk Import** to assign a batch of students to that specific exam slot. The backend automatically matches student rows against existing database records using any of the identifier columns below:

Header Name	Req?	Lookup Matching Behavior & Example
email	ANY 1	Primary lookup by exact student email (e.g., student1@bitsathy.ac.in)
enrollment_no	ANY 1	Matches university enrollment number (e.g., 737622CS101)
roll_no	ANY 1	Matches student roll number (e.g., 22CS101)
user_id	ANY 1	Matches internal numeric User ID (e.g., 15)
full_name	ANY 1	Fallback match by exact full name (if unique among students)

D. Question Bank Bulk Import (AdminQuestions_new.jsx):

Used to import dozens of coding and interactive Flutter UI problems simultaneously.

Header Name	Req?	Format / Example
level	YES	Level code matching DB (e.g., 1A, 2B, 3C)
title	YES	Question title (e.g., <i>Login Form Validation</i>)

Header Name	Req?	Format / Example
description	YES	Detailed Markdown instructions & rules
starter_code	YES	Dart string OR JSON object map: { <code>"lib/main.dart": "import..."</code> }
ui_required_widgets	NO	JSON Array: [<code>"Scaffold", "AppBar", "ListView"</code>]
required_packages	NO	JSON Array: [<code>"provider", "http", "google_fonts"</code>]
mock_api_route	NO	Endpoint string: <code>/api/mock/users</code>
mock_api_response	NO	JSON payload: { <code>"users": [{<code>"id": 1</code>}]</code> }
custom_test_code	NO	Complete testWidgets Dart test suite
is_active	NO	Either true or 1 to make active upon import

■ **CRITICAL TIP FOR CSV MULTI-LINE CODE & STRINGS:** When writing code snippets, JSON maps, or multi-line descriptions inside spreadsheet cells (Excel/Numbers/Google Sheets), wrap the entire cell contents in double quotes " and press Alt+Enter (Windows) or Option+Return (Mac) for line breaks. Escape inner double quotes by doubling them (e.g., `"lib/main.dart"`).

5. Test Slot & Schedule Configuration

To administer an exam, navigate to **Test Schedules / Slots** (AdminTestSlots.jsx) and click **+ Create Schedule**.

- **Schedule Details:** Set Schedule Name (e.g., *Batch 2026 Mid-Term*), Start/End Date & Time, and Duration Minutes.
- **Staff Assignment:** Assign specific staff members to three distinct operational roles:
 - **Live Teacher:** Responsible for live classroom supervision and addressing student queries.
 - **Code Reviewer Teacher:** Assigned to evaluate algorithm/logic tests (AdminSubmissions.jsx).
 - **UI Reviewer Teacher:** Assigned to evaluate interactive Flutter UI tests (AdminManualGrading.jsx).

Once created, registered students will see the assessment unlock on their dashboard at the designated start time.

6. Student Portal Rules & Execution Environment

When students start an assessment, they enter a monitored, high-performance coding environment tailored for Flutter development.

- **Dark Monaco Editor:** Configured strictly in Dark Theme (vs-dark) for optimal syntax highlighting and readability.
- **Interactive Live Preview:** Clicking **Run Preview** compiles code server-side and opens an interactive HTML5 viewport on the right half of the screen. Students can click buttons, type text, scroll lists, and test animations in real-time!
- **Submission Mechanics:** Clicking **Submit** records the multi-file code map and freezes the score. If a student disconnects, session timers persist server-side while code drafts are restored from browser storage upon

reconnection.

7. Grading & Result Analysis (With Excel Export)

Grading combines **Automated AI/Test Assertions (50%)** and **Teacher Manual Review (50%)**. Both code review pages now feature instant searchable filtering and one-click Excel (.xlsx) spreadsheet exporting!

A. Coding Test Code Review (AdminSubmissions.jsx):

- **Searchable Filtering:** Use the search bar to instantly search across Student Name, Email, Roll Number, Problem Title, or Slot Name.
- **Status Filter:** Filter submissions by ALL, PASS, or FAIL.
- **Export XLSX Spreadsheet:** Click the green **Export XLSX** button. Instantly downloads an Excel sheet containing: S.No, Submission ID, Student Name, Roll Number, Email, Test Slot, Problem ID, Status, Auto Score (%), Match Percent, Code Length, and Timestamp!

B. UI Test Code Review (AdminManualGrading.jsx):

- **Searchable Filtering:** Use the newly added search input bar to instantly filter UI submissions by Student Name, Roll Number, Email, Problem Title, or Slot Name.
- **Live Interactive Preview Review:** Select any student submission and click the **Live Preview & Mockup** tab. Teachers can interact with the student's compiled Flutter web app live inside the iframe to check animations, responsiveness, and button behaviors!
- **Student Code Inspection:** Click the **Student Code** tab to inspect all submitted files (lib/main.dart, models, widgets).
- **Assigning Manual Grades & Feedback:** Enter a Manual Score (0–100) and constructive feedback. The system auto-computes the final score:
$$\text{Final Score} = (\text{Automated Score} \times 0.5) + (\text{Manual Score} \times 0.5)$$
- **Export XLSX Spreadsheet:** Click the green **Export XLSX** button. Instantly downloads an Excel sheet containing: S.No, Submission ID, Student Name, Email, Roll Number, Test Slot, Problem Title, Auto Score (50%), Manual Score (50%), Final Combined Score (%), Teacher Feedback, Status, and Code Files Count!